Erasmus Mundus Joint Master Artificial Intelligence for Sustainable Societies (AISS)

Topics on Machine Learning and its Applications

Exploratory data analysis

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Exploratory data analysis

- Introduction
- Types of exploratory data analysis
- Missing data analysis
- Case study of EDA

Exploratory data analysis

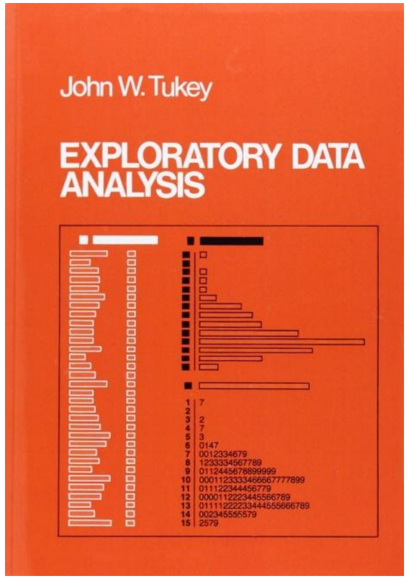
Introduction

- **What is Exploratory Data Analysis (EDA)?**
  - EDA is the first step in any data analysis project
  - It involves exploring, examining and visualizing data
- **Why is EDA important?**
  - **Summarize main characteristics** – Get a sense of the data's center, spread, and shape
  - **Identify patterns** – Discover relationships, trends, and anomalies
  - **Gain insights** – Uncover underlying structure before formal modeling and evaluate whether ML techniques are necessary

Exploratory data analysis

Introduction

- **Origins**
  - Introduced by John Tukey in the 1970s
  - Combines statistical tools and data discovery



- Conceptually, Tukey distinguished between
  - **Exploratory analysis** – Discover patterns without strong prior assumptions
  - **Confirmatory analysis** – Test predefined hypotheses using formal statistical inference

Exploratory data analysis

Introduction

- **Classical analysis**
  - Formulate a hypothesis
  - Confirms a hypothesis
  - Deductive reasoning
- **Exploratory data analysis**
  - Empirical and inspection
  - Generates hypotheses from the data
  - Inductive reasoning
- **Purposes**
  - Determine key questions
  - Assess data objectively
  - Describe data
  - Understand findings
  - Prepare for complex analyses
- **Applications**
  - Data analysis
  - Stakeholder engagement
  - Model selection

Exploratory data analysis
Introduction

In CRISP-DM **EDA** corresponds to **Data understanding**

- The **data understanding** phase begins with initial data collection and then dives into exploring and assessing that data
- This involves **getting familiar with its structure**, identifying potential quality issues, uncovering preliminary insights, and pinpointing interesting subsets to form hypotheses for hidden information
- This phase allows analysts to **grasp the data's characteristics** and validate its reliability as a foundation for drawing valid inferences and building robust models
- A significant portion of time should be dedicated to this phase, as a deep understanding of the data is essential for mitigating risks and making informed business decisions based on subsequent analysis

Types of Exploratory data analysis

Major EDA techniques landscape

Method	**Scope**	
	Single variable	**Multiple variables**
Statistical measures	Univariate non-graphical	Multivariate non-graphical
Visualization techniques	Univariate graphical	Multivariate graphical

Types of Exploratory data analysis

Univariate non-graphical

- **Focus**
  - Single variable analysis
- **Major techniques**
  - Statistical measures
    - Central tendency (e.g. mean, median, mode)
    - Spread (e.g. range, MAD, variance, standard deviation, range)
    - Distribution (e.g. skewness, outliers)
- **Purpose**
  - To summarize and describe a single variable's distribution

Types of Exploratory data analysis

Univariate non-graphical

- **Measures of central tendency**
  - For interval or ratio level data, one measure of center is the **mean**
  - The *population mean* is denoted by μ
  - The *sample mean* intended to estimate it is denoted by $\bar{x}$
  - Assuming the population has size N, a sample has size n, and x spans across all available examples in the population or sample, as appropriate, these means are calculated by:

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N} \qquad \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

  - The arithmetic mean is a **non-robust** estimator, as it can be substantially influenced by extreme observations

Types of Exploratory data analysis

Univariate non-graphical

- **Measures of central tendency**
  - The **median**, denoted by is the middle value of a set when it is written in order
  - In the case of an even number of data values (and thus no exact middle), it is the average of the middle two data values

1, 3, 3, **6**, 7, 8, 9

Median = $\underline{\underline{6}}$

1, 2, 3, **4**, **5**, 6, 8, 9

Median = (4 + 5) ÷ 2

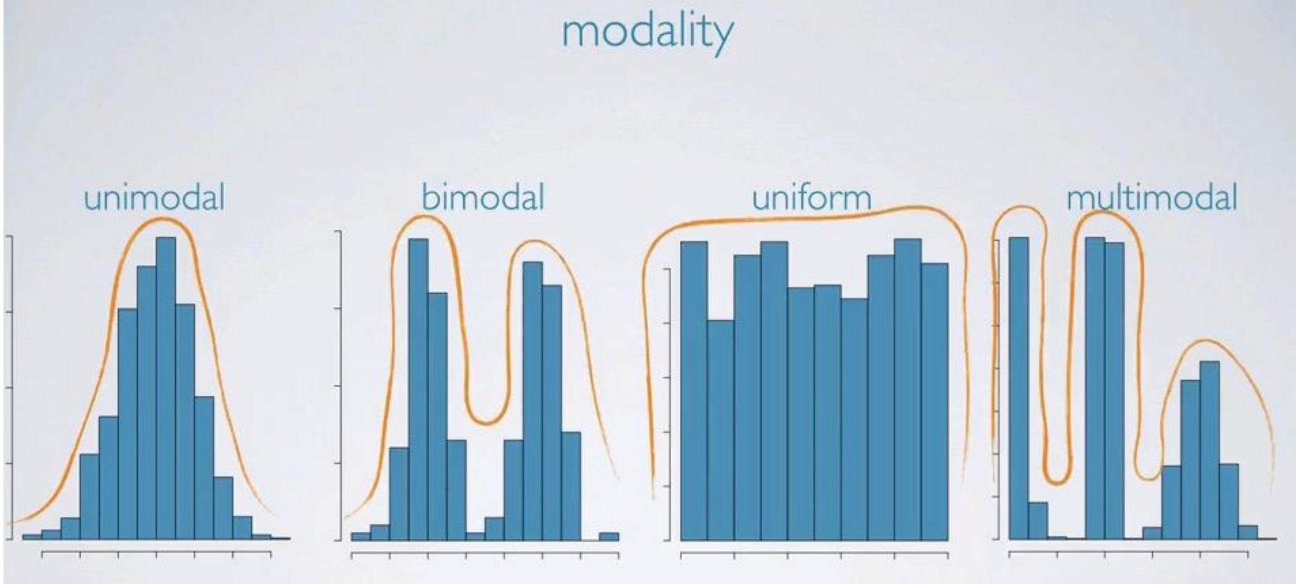
= $\underline{\underline{4.5}}$

  - It is not affected by the presence of extreme values in the data set, so it is considered more **robust** than the mean
  - Unlike the mean, it can sometimes even suggest a central value for categorical ordinal data

Types of Exploratory data analysis

Univariate non-graphical

- **Measures of central tendency**
  - The **mode** is the most frequent data value in the population or sample
  - There can be more than one mode, although in the case where there are no repeated data values, it is said that *there is no mode*

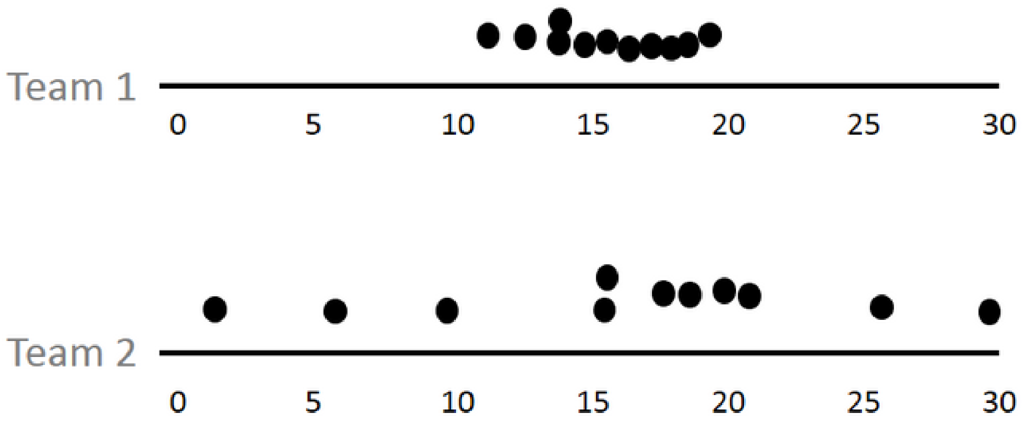


  - Modes can be used both for numerical and categorical data

Types of Exploratory data analysis

Univariate non-graphical

- **Measures of spread**
  - Considering the age distribution of 2 teams of players

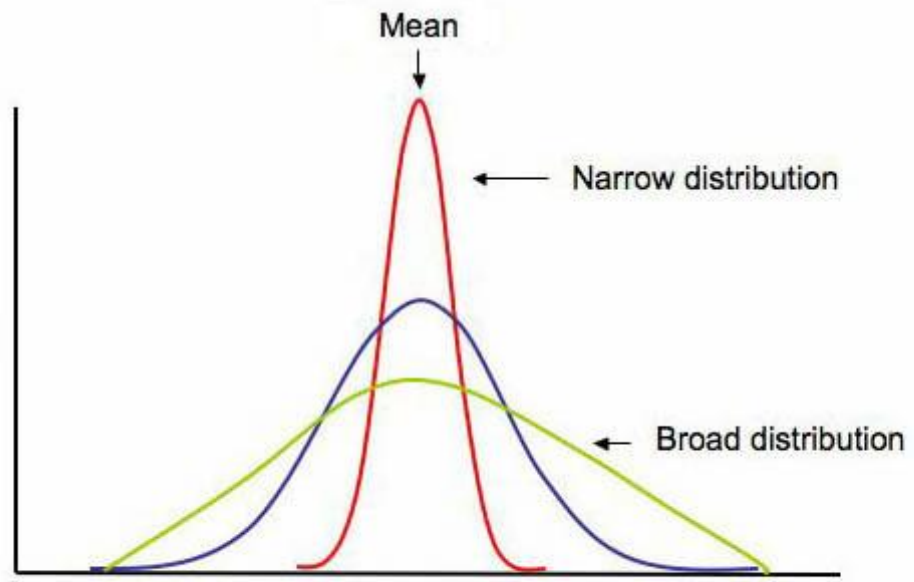


  - Mode = 14.1, Median = 15 and Mean = 15
  - To adequately describe a distribution, it may need more information than the measures of central tendency

Types of Exploratory data analysis

Univariate non-graphical

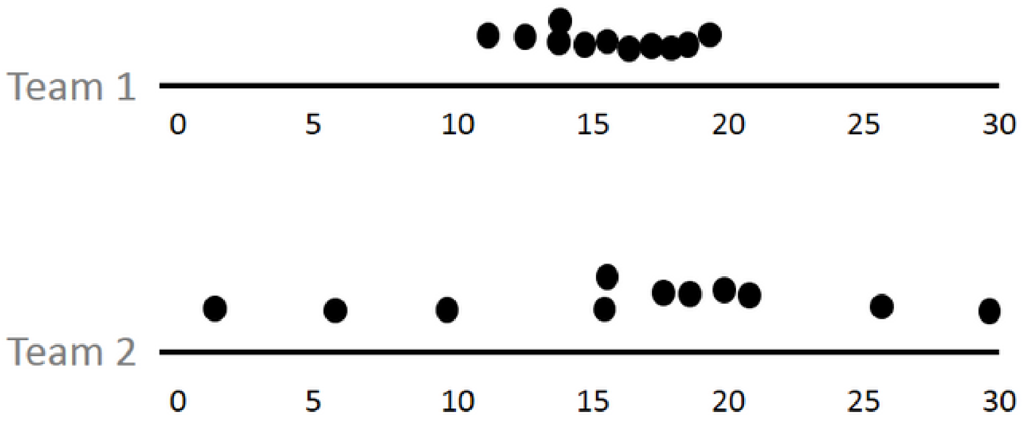
- **Measures of spread**
  - The **range** is the difference between the highest and lowest values of a distribution, although it is often reported by simply listing the minimum and maximum values seen
  - It is strongly affected by extreme values present in the distribution



Types of Exploratory data analysis

Univariate non-graphical

- **Measures of spread**



  - Range (Team1) = 19.3 – 10.8 = 8.5
  - Range (Team2) = 30.0 – 1.0 = 29.0
  - As ranges takes only the count of extreme values sometimes it may not give a good impact on variability

Types of Exploratory data analysis

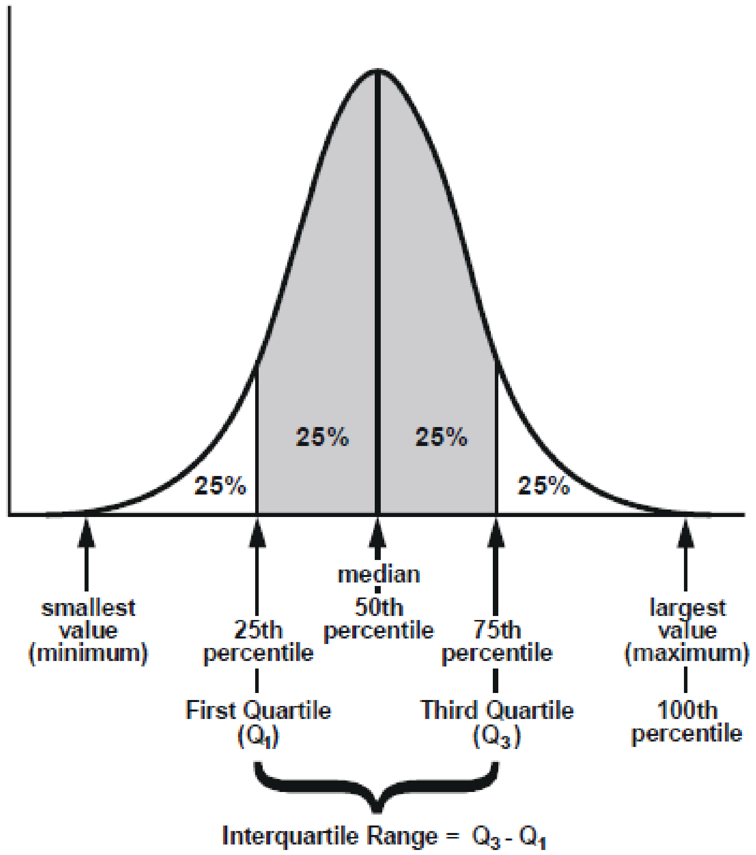
Univariate non-graphical

- **Measures of spread**
  - The **InterQuartile Range** (IQR) is a better measure of dispersion than range because it leaves out the extreme values
  - It equally divides the distribution into four equal parts called **quartiles**
    - First 25% is the 1st quartile (Q_1)
    - Middle one is the 2nd quartile (Q_2, which is also the median)
    - Last one is the 3rd quartile (Q_3)
  - When the data are sorted, the IQR is simply the range of the middle half of the data

$$IQR = Q_3 - Q_1$$

Types of Exploratory data analysis
Univariate non-graphical

- **Measures of spread**



Range, Interquartile Range and Box Plot https://makemeanalyst.com/explore-your-data-range-interquartile-range-and-box-plot/

Types of Exploratory data analysis

Univariate non-graphical

- **Measures of spread**
  - The **Mean Absolute Deviation** (MAD), is the average distance to the mean
  - The distance between two values x_1 and x_2 is given by the absolute value of their difference $|x_1 - x_2|$, so the distance between a value x and the mean of the population μ would be $|x - \mu|$
  - To find the average of this distance, the differences are added over the population and divided by the number of examples in the population, N:

$$MAD = \frac{\sum_{i=1}^{N} |x_i - \mu|}{N}$$

  - For a sample, the procedure is the same

Types of Exploratory data analysis

Univariate non-graphical

- **Measures of spread**
  - The **variance** (σ^2) is the average squared distance between population values
  - The **standard deviation** (σ) is the square root of the variance

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} \quad \text{and} \quad \sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

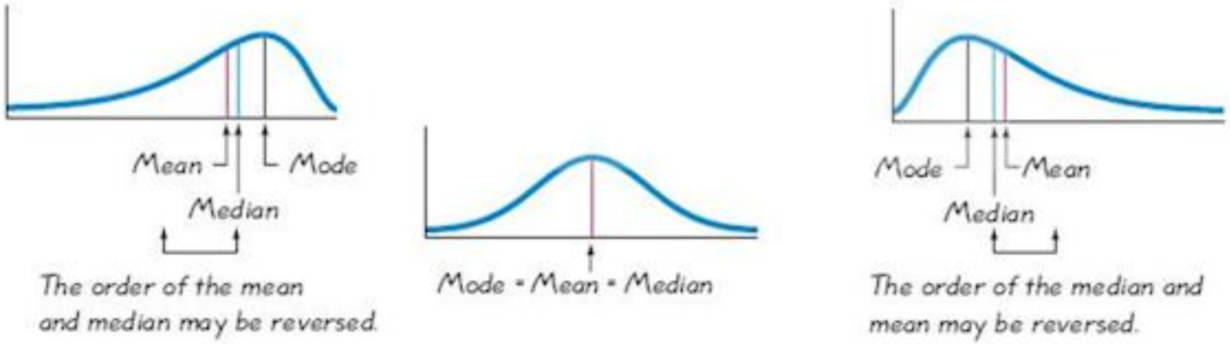
  - Both can exaggerate the contributions to the spread of the population made by values far from the mean
  - When dealing with a sample the *Bessel's Correction* is used (https://mathcenter.oxford.emory.edu/site/math117/besselCorrection/)

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} \quad \text{and} \quad s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Types of Exploratory data analysis

Univariate non-graphical

- **Distribution**
  - The presence of **skewness** can affect where the measures of central tendency are located relative to one another



Shape, Center, and Spread of a Distribution: https://mathcenter.oxford.emory.edu/site/math117/shapeCenterAndSpread/

  - When significant skewness is present, the mean and median end up in different places

Types of Exploratory data analysis

Univariate non-graphical

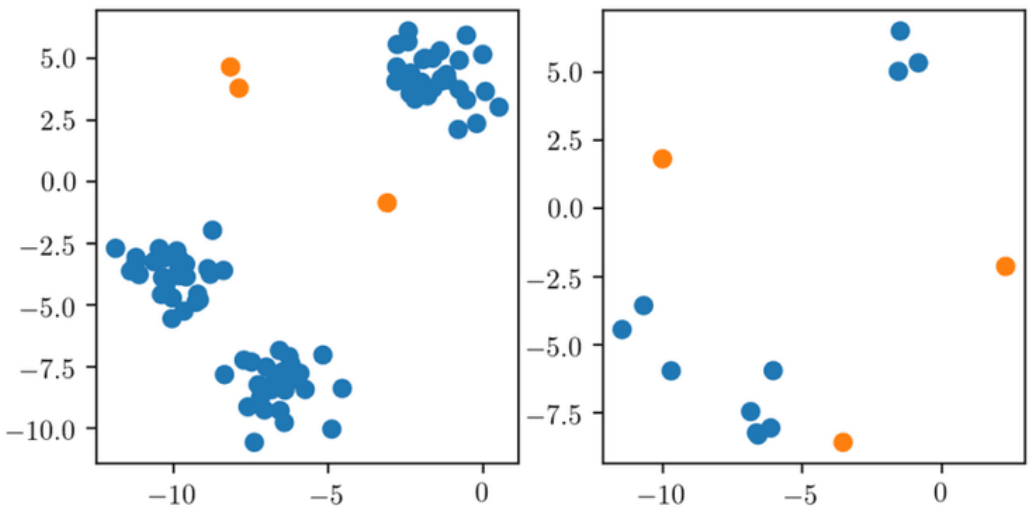
- **Distribution**
  - If the mean and median are far enough apart, it is possible to determine if an observed skewness is significant
  - The **Pearson's Skewness Index** (I) is defined as:

$$I = \frac{3(\bar{x} - Q_2)}{s}$$

  - The interpretation is that the data **is significantly skewed** if $|I| \geq 1$

Types of Exploratory data analysis
Univariate non-graphical

- **Outliers**
  - An outlier is a data value significantly far removed from the main body of a data set
    - Also considered extreme values that deviate significantly from other data points
    - Outliers can have a disproportionate impact on the results of statistical models, especially in algorithms sensitive to large values



- Techniques for outlier detection:
  - **Visualization** – Box plots and scatter plots are often used to identify outliers visually
  - **Statistical methods**
    - Z-Score
    - Interquartile Range (IQR)

Types of Exploratory data analysis

Univariate non-graphical

- **Outliers**
  - The **Z-Score** measures how many standard deviations an example is from the mean

$$Z = \frac{X - \mu}{\sigma}$$

  - Detecting outliers with the **Z-Score**:
    - *An example is an outlier if*

$$Z > 3 \quad \vee \quad Z < -3$$

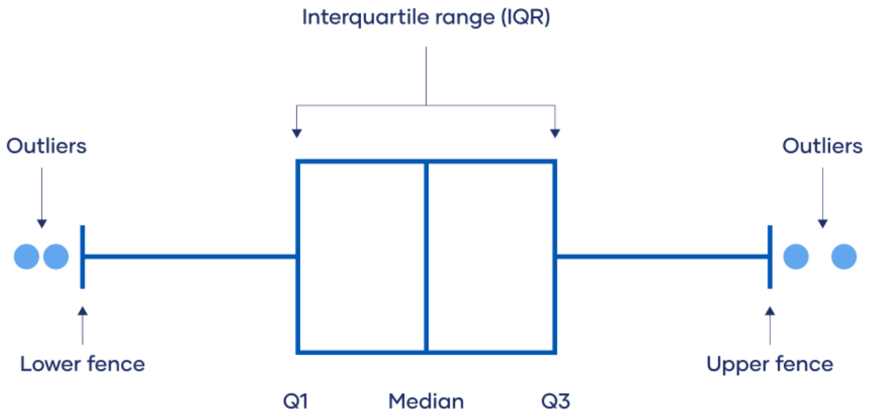
    - A positive Z-Score ($Z > 0$) means the example is above the mean
    - A negative Z-Score ($Z < 0$) means the example is below the mean
    - A Z-Score of 0 means the example is exactly at the mean

Types of Exploratory data analysis

Univariate non-graphical

- **Outliers**
  - Detecting outliers with the **InterQuartile Range**:
    - *An example is an outlier if*

$$X < (Q1 - 1.5\, IQR\,) \;\vee\; X > (Q3 + 1.5\, IQR\,)$$



How to find outliers: https://www.scribbr.com/statistics/outliers/

Types of Exploratory data analysis

Multivariate non-graphical

- **Focus**
  - Analyzing multiple variables simultaneously

- **Major techniques**
  - Cross-tabulation – For a low number of variables, creates a table to show the relationship between different levels of variables
  - Covariance – Measures the direction of the relationship between variables
  - Correlation – Measures the strength and direction of the relationship between variables

- **Purpose**
  - Understand the relationships between multiple variables

Types of Exploratory data analysis

Multivariate non-graphical

- **Cross-tabulation**
  - For a low number of categories, creates a table to show the relationship between different levels of two variables
  - Example: “Lung cancer” vs “Smoking habits”

		Has lung cancer?		Total
		Yes	No	
Is a smoker?	Yes	300	2700	3000
	No	100	6900	7000
Total		400	9600	10000

  - This table suggests a potential relationship, with a higher proportion of smokers developing lung cancer compared to non-smokers

Types of Exploratory data analysis

Multivariate non-graphical

- **Positive covariance and correlation**
  - For a low number of categories, creates a table to show the relationship between different levels of two variables
  - Example: “Ice creams sales” vs “Temperature”

		Temperature		Total
		Low	High	
Sales performance	Low	10	2	12
	High	3	15	18
Total		13	17	30

  - This table shows that higher ice cream sales tend to occur with higher temperatures, suggesting a positive relationship

Types of Exploratory data analysis

Multivariate non-graphical

- **Negative covariance and correlation**
  - For a low number of categories, creates a table to show the relationship between different levels of two variables
  - Example: “Hot chocolate sales” vs “Temperature”

		Temperature		Total
		Low	High	
Sales performance	Low	2	10	12
	High	15	3	18
Total		17	13	30

  - This table demonstrates that higher hot chocolate sales are more frequent with lower temperatures, indicating a negative relationship

Types of Exploratory data analysis

Multivariate non-graphical

- The previous tables categorize variables (temperature and sales) into *high* and *low* for illustrative purposes in this format

- The actual analysis would involve calculating covariance and correlation coefficients using the precise numerical values

$$Cov(x_1, x_2) = \frac{1}{N}\sum_{i=1}^{N}(x_{1i} - \mu_1)(x_{2i} - \mu_2)$$

- Correlation does not necessarily equal causation!

Types of Exploratory data analysis

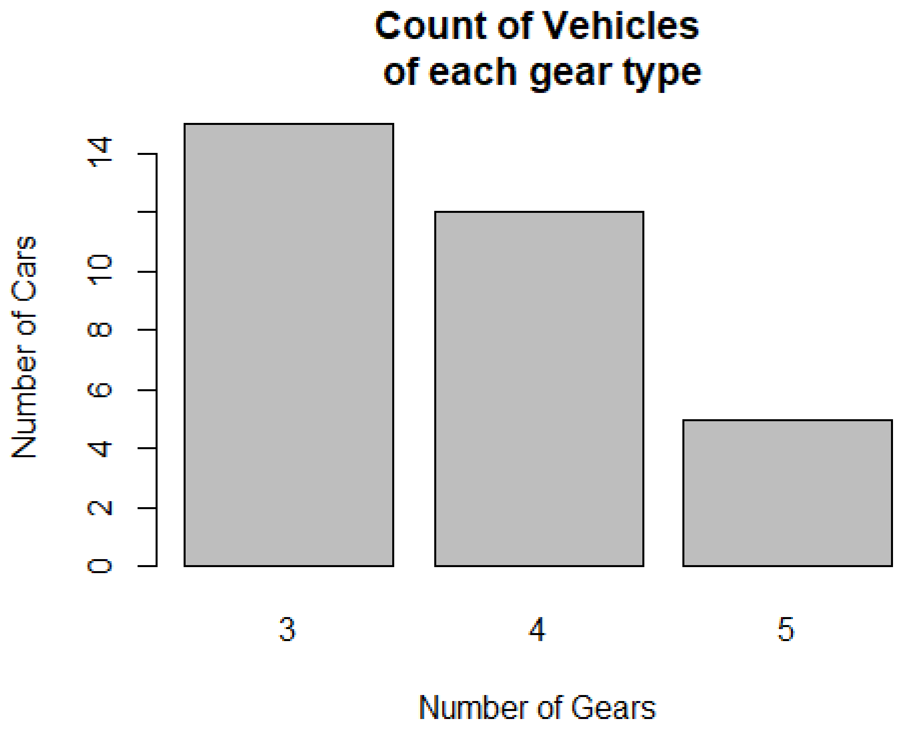
Univariate graphical

- **Focus**
  - Visualizing the distribution of a single variable
- **Major techniques**
  - Bar plots
  - Histograms
  - Box plots
- **Purpose**
  - To visually describe a single variable's distribution

Types of Exploratory data analysis

Univariate graphical

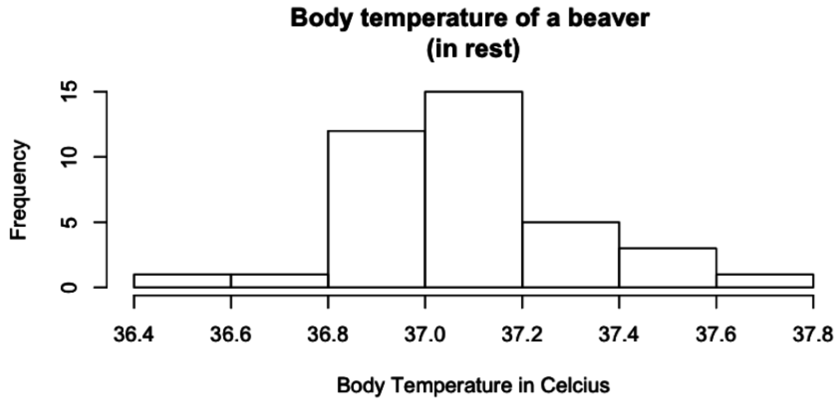
- **Bar plot** – categorical variables

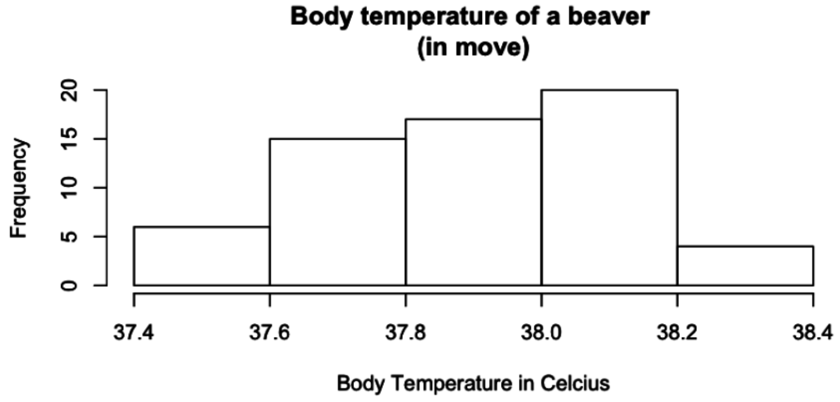


Barplots, Histograms and Boxplots:
https://sustainabilitymethods.org/index.php/Barplots,_Histograms_and_Boxplots

Types of Exploratory data analysis
Univariate graphical

- **Histograms** – continuous variables





Barplots, Histograms and Boxplots:
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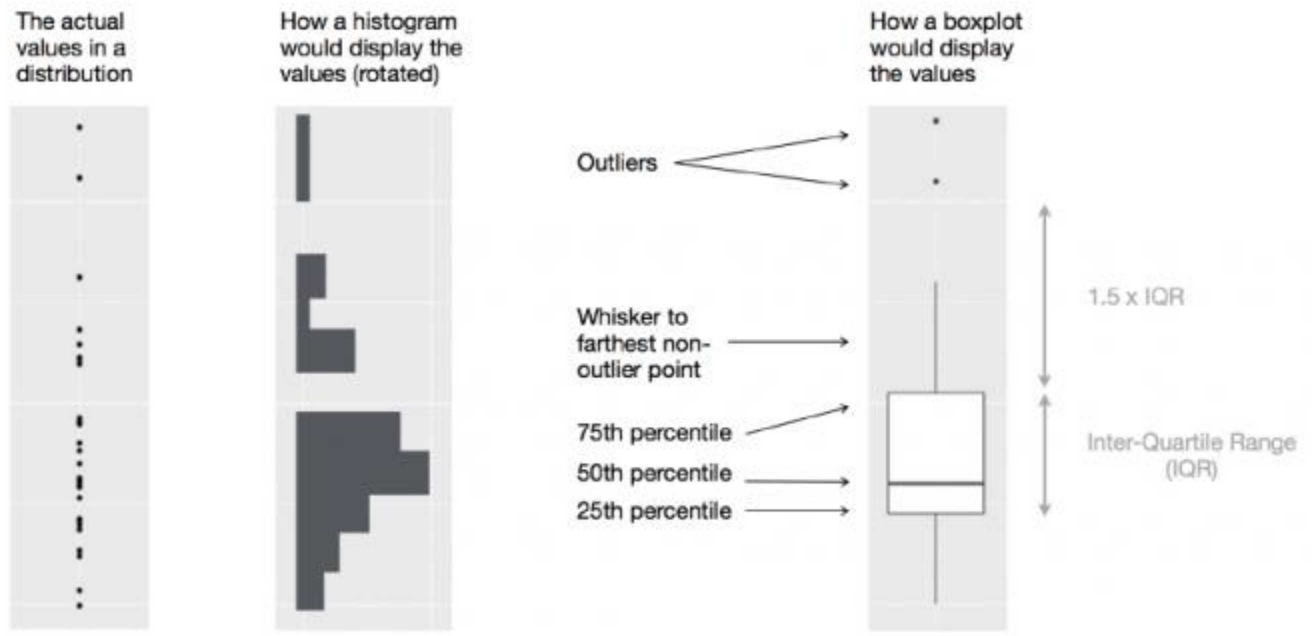
Types of Exploratory data analysis

Univariate graphical

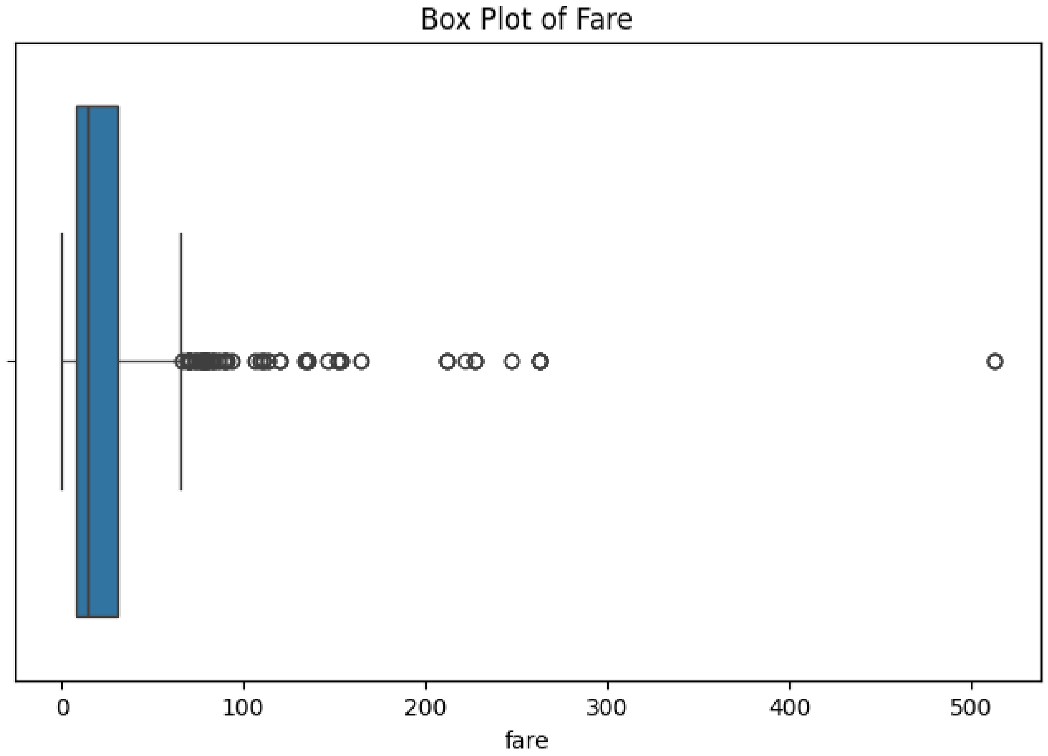
- **Box-plots** – continuous variables



Barplots, Histograms and Boxplots: https://sustainabilitymethods.org/index.php/Barplots,_Histograms_and_Boxplots

Types of Exploratory data analysis
Univariate graphical

- **Skewness and outliers** using a **boxplot**
  - Variable *Fare* in the Titanic dataset



Types of Exploratory data analysis

Multivariate graphical

- **Focus**
  - Simultaneous visualization of the distribution of a multiple variables

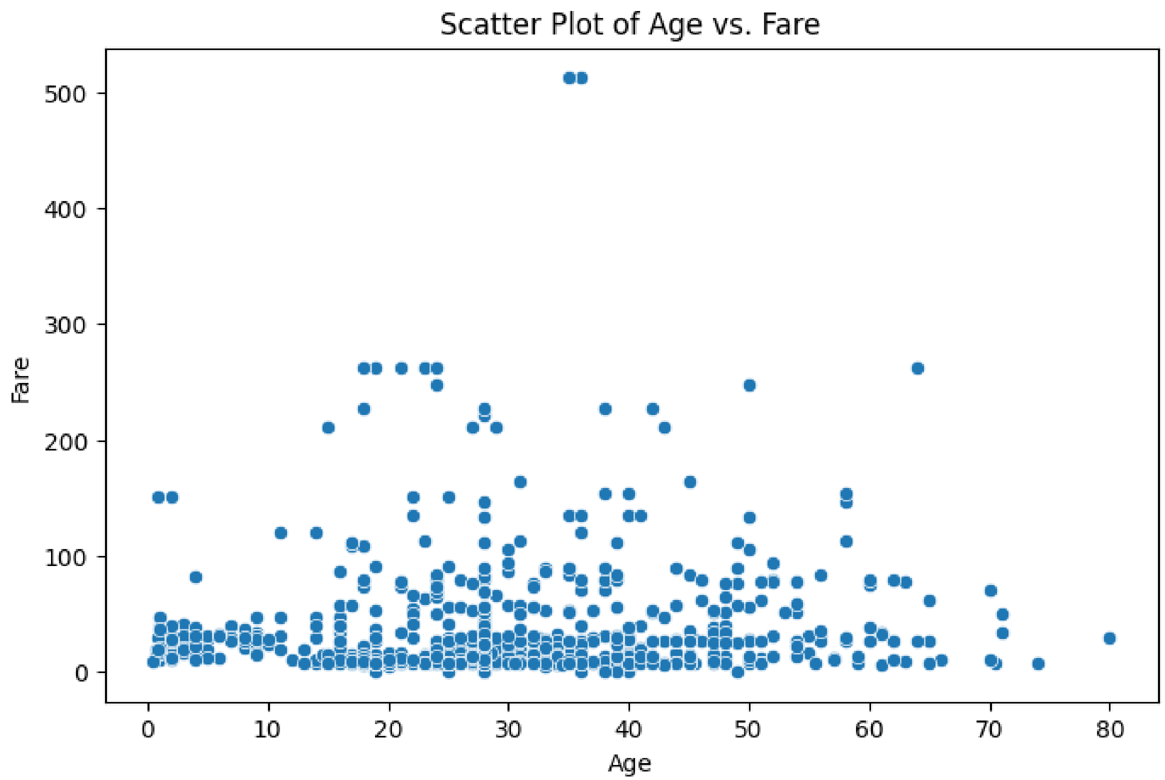
- **Major techniques**
  - Scatterplots
  - Grouped bar plot
  - Heat maps
  - Multivariate histograms

- **Purpose:**
  - Analyze relationships between multiple variables

Types of Exploratory data analysis

Multivariate graphical

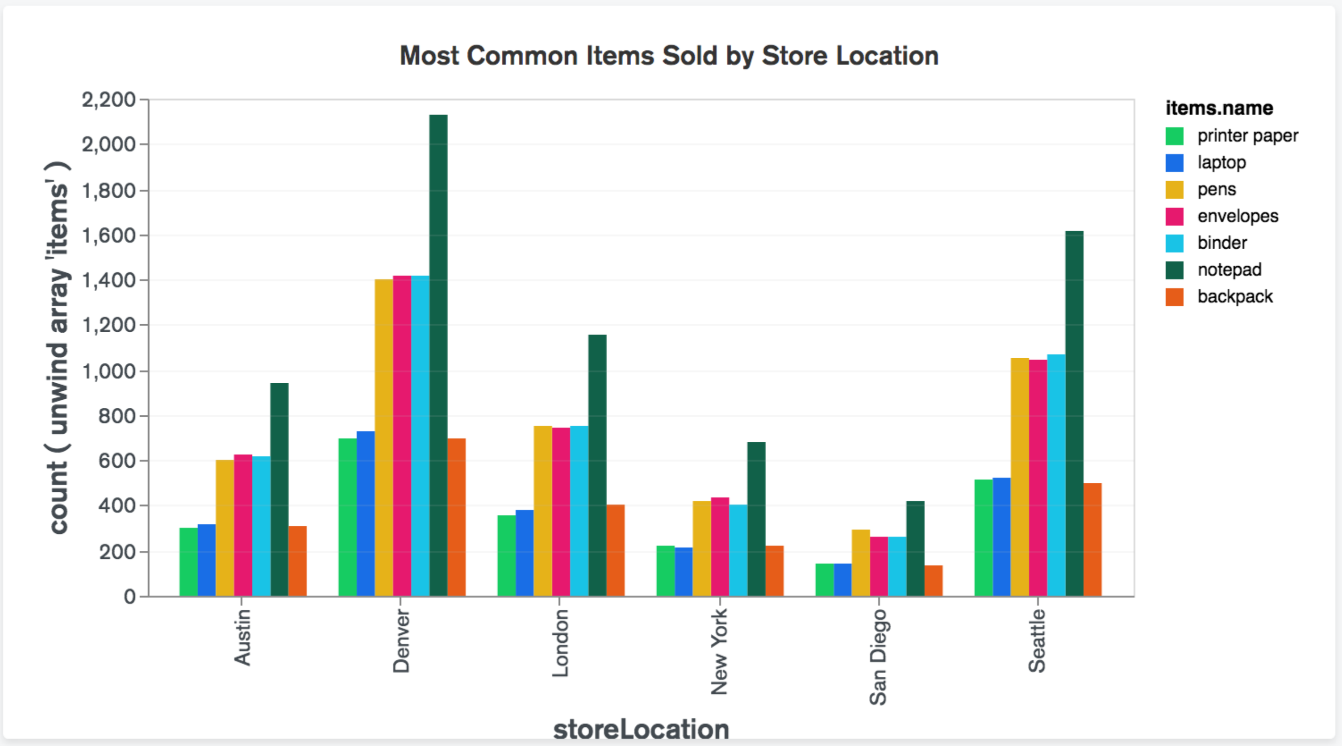
- **Scatterplot**
  - Variables *Fare* and *Age* in the Titanic dataset



Types of Exploratory data analysis

Multivariate graphical

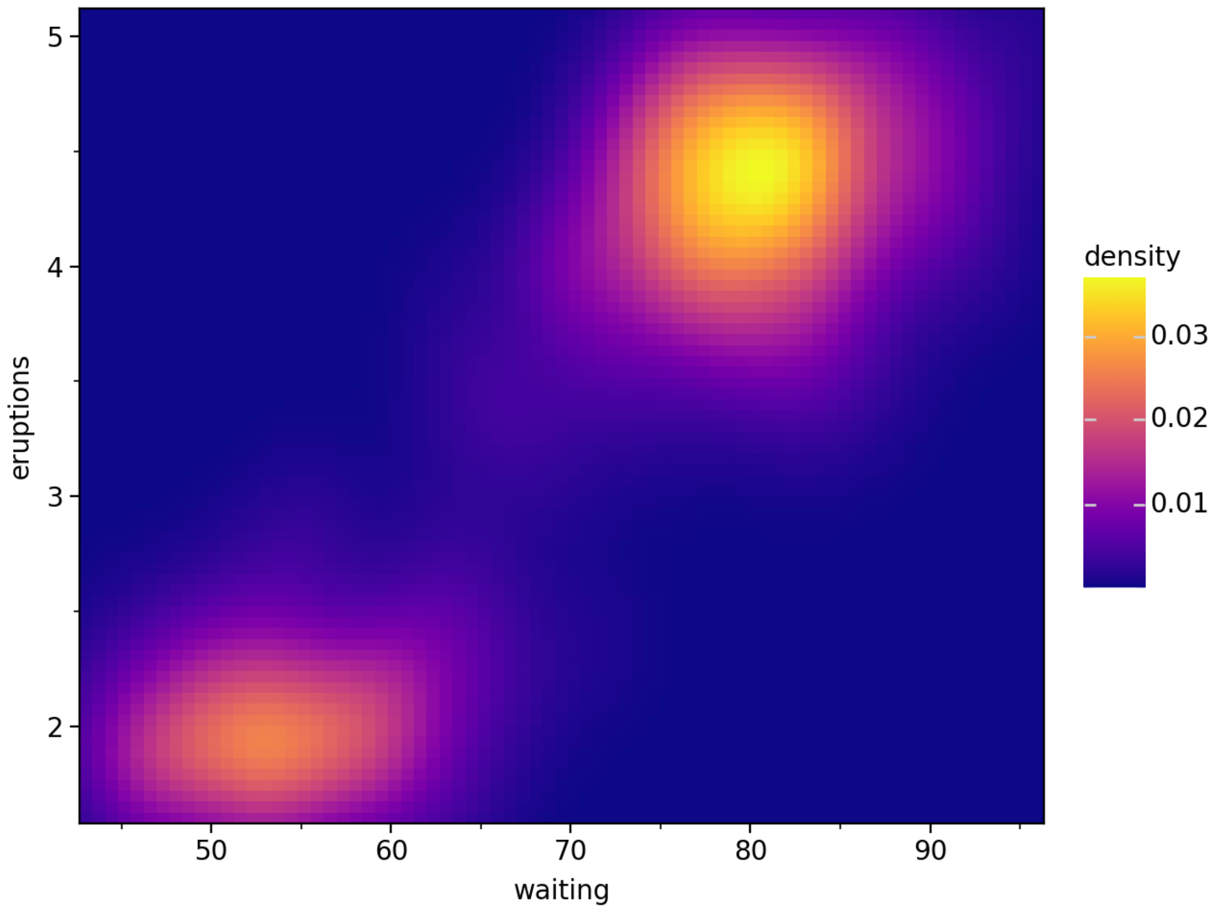
- **Grouped bar plot**



Types of Exploratory data analysis

Multivariate graphical

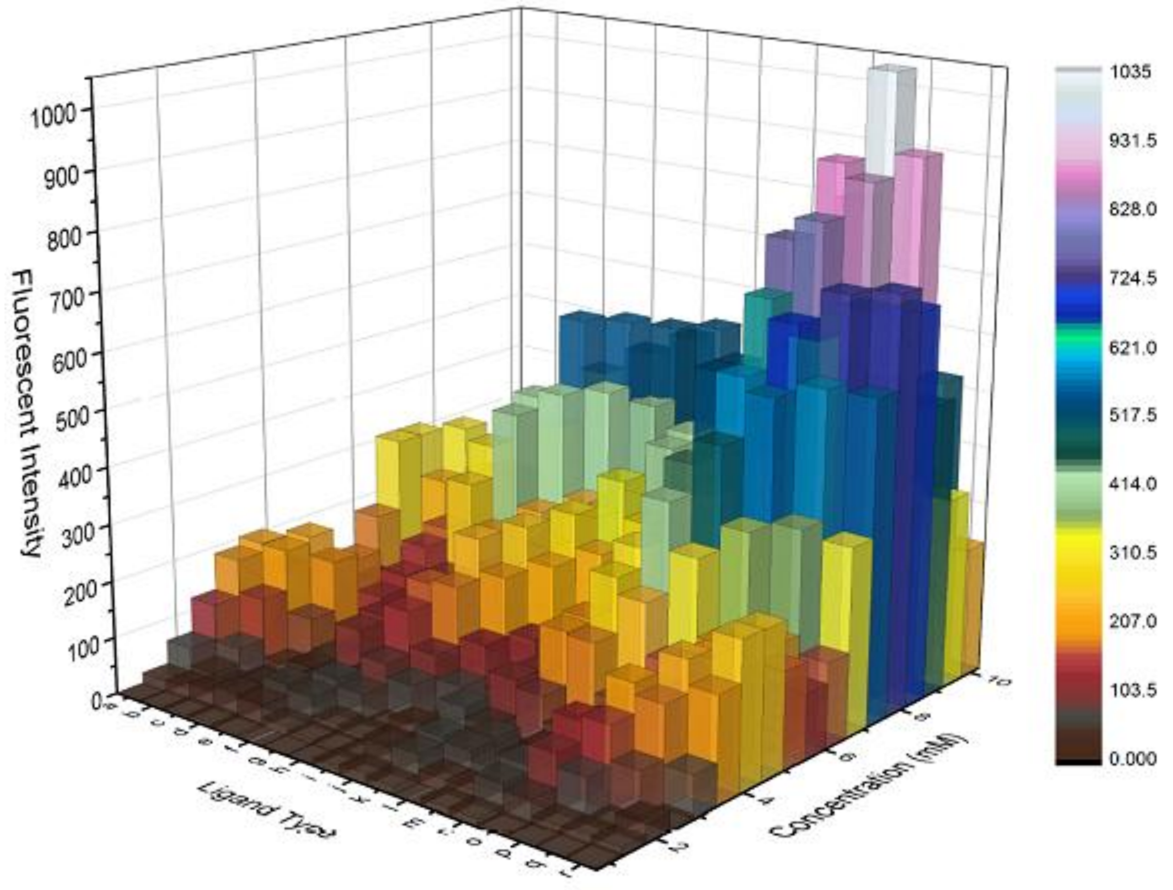
- **Heat map** – continuous or discrete variables



Types of Exploratory data analysis

Multivariate graphical

- **Multivariate histograms** – continuous or discrete variables



Types of Exploratory data analysis

Missing data analysis

- Key aspects of data quality
  - Accuracy
  - Completeness
  - Consistency

- Challenges with data quality
  - Inaccurate, incomplete, and inconsistent data are **common** in real-world databases
  - Missing data can impact decision-making, analysis, and model performance

Types of Exploratory data analysis

Missing data analysis

- Causes of **inaccurate data**
  - Faulty data collection instruments – Equipment errors can lead to incorrect data
  - Human/Computer errors – Mistakes during data entry or automated processing
  - Disguised missing data – Users may intentionally input incorrect or default values (e.g., "January 1" as a placeholder for birthdays)
  - Transmission errors – Data can become corrupted during transmission due to issues such as limited buffer size
  - Naming inconsistencies – Mismatched naming conventions or inconsistent data codes lead to errors
  - Inconsistent formats – Different formats for data input (e.g., date formats) cause inaccuracies

Types of Exploratory data analysis

Missing data analysis

- How to identify missing data visually?

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0.0	A/5 21171	7.2500		S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1			71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female		0	0.0	STON/O2. 3101282	7.9250		S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0.0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0.0	373450	8.0500		S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0.0	211536	13.0000		S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0.0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female		1	2.0	W./C. 6607	23.4500		S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0.0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0.0	370376	7.7500		Q

891 rows × 12 columns

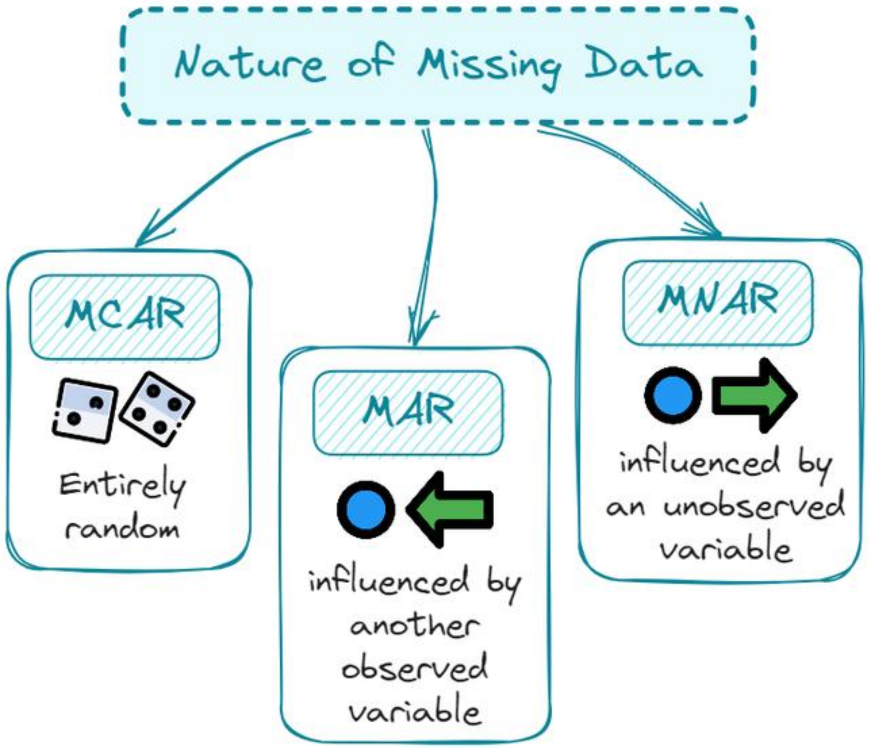
	Survived	Age	Fare
0	1.0	22.0	7.250
1	NaN	NaN	NaN
2	NaN	26.0	7.925
3	NaN	NaN	53.100
4	0.0	35.0	8.050
...	...	...	...
886	0.0	27.0	13.000
887	1.0	19.0	30.000
888	0.0	NaN	23.450
889	1.0	26.0	30.000
890	0.0	32.0	7.750

891 rows × 3 columns

Types of Exploratory data analysis

Missing data analysis

- **Types of missing data**
  - Missing Completely At Random (MCAR)
  - Missing At Random (MAR)
  - Missing Not At Random (MNAR)



Concepts of MCAR, MAR and MNAR: https://stefvanbuuren.name/fimd/sec-MCAR.html

Types of Exploratory data analysis

Missing data analysis

- **Missing Completely At Random (MCAR)**
  - Data are MCAR when the probability of missingness does not depend on any variable, observed or unobserved
  - The missing values occur purely by chance
  - Examples:
    - A sensor randomly fails
    - Some questionnaires are lost due to a printing error
    - A lab sample is accidentally destroyed
  - Consequence:
    - The observed data are a random subsample
    - Complete-case analysis remains unbiased
    - Statistical power is reduced, but no systematic bias is introduced

Types of Exploratory data analysis

Missing data analysis

- **Missing At Random (MAR)**
  - Data are MAR when missingness depends only on observed variables, not the missing value itself
  - Once we control for observed values, the missingness process no longer depends on the missing value
  - Examples:
    - Suppose in a questionnaire “income” is missing more often on younger respondents
    - If “age” is observed then missingness depends on age, not in the income value itself
  - Consequence:
    - Ignoring missingness may introduce bias
    - However, methods like multiple imputation or maximum likelihood can produce unbiased estimates if the model includes the relevant observed predictors

Types of Exploratory data analysis

Missing data analysis

- **Missing Not At Random (MNAR)**
  - Data are MNAR when missingness depends on the unobserved value itself, even after accounting for observed variables
  - The reason for missingness is directly related to the missing value.
  - Examples:
    - High-income individuals refuse to report “income”
    - Students with very low grades avoid answering a survey about academic performance
    - Patients with severe symptoms drop out of a study
  - Consequence:
    - Standard imputation methods assuming MAR may produce biased results
    - Modeling the missingness mechanism explicitly is required
    - Strong assumptions are unavoidable

Case study of EDA

Titanic dataset

Variable	Type	Description
survived	Integer numeric	Indicates whether the passenger survived (1) or not (0)
pclass	Integer numeric	Ticket class (1, 2 or 3)
sex	Nominal categorical	Passenger’s gender (male, female)
age	Continuous numeric	Age at the time of the voyage
sibsp	Discrete numeric	Number of siblings/spouses aboard
parch	Discrete numeric	Number of parents/children aboard
fare	Continuous numeric	Ticket price (in £)
embarked	Nominal categorical	Port of embarkation initial (C, Q, S)
class	Ordinal categorical	Textual equivalent of ticket class (First, Second, Third)
who	Nominal categorical	Passenger category (man, woman, child)
adult_male	Boolean	Indicates whether the passenger is an adult male (T/F)
deck	Nominal categorical	Deck level, with missing values (NaN)
embark_town	Nominal categorical	Port of embarkation (Cherbourg, Queenstown, Southampton)
alive	Nominal categorical	Textual equivalent of survived (yes, no)
alone	Binary	Indicates whether the passenger was travelling alone (T/F)

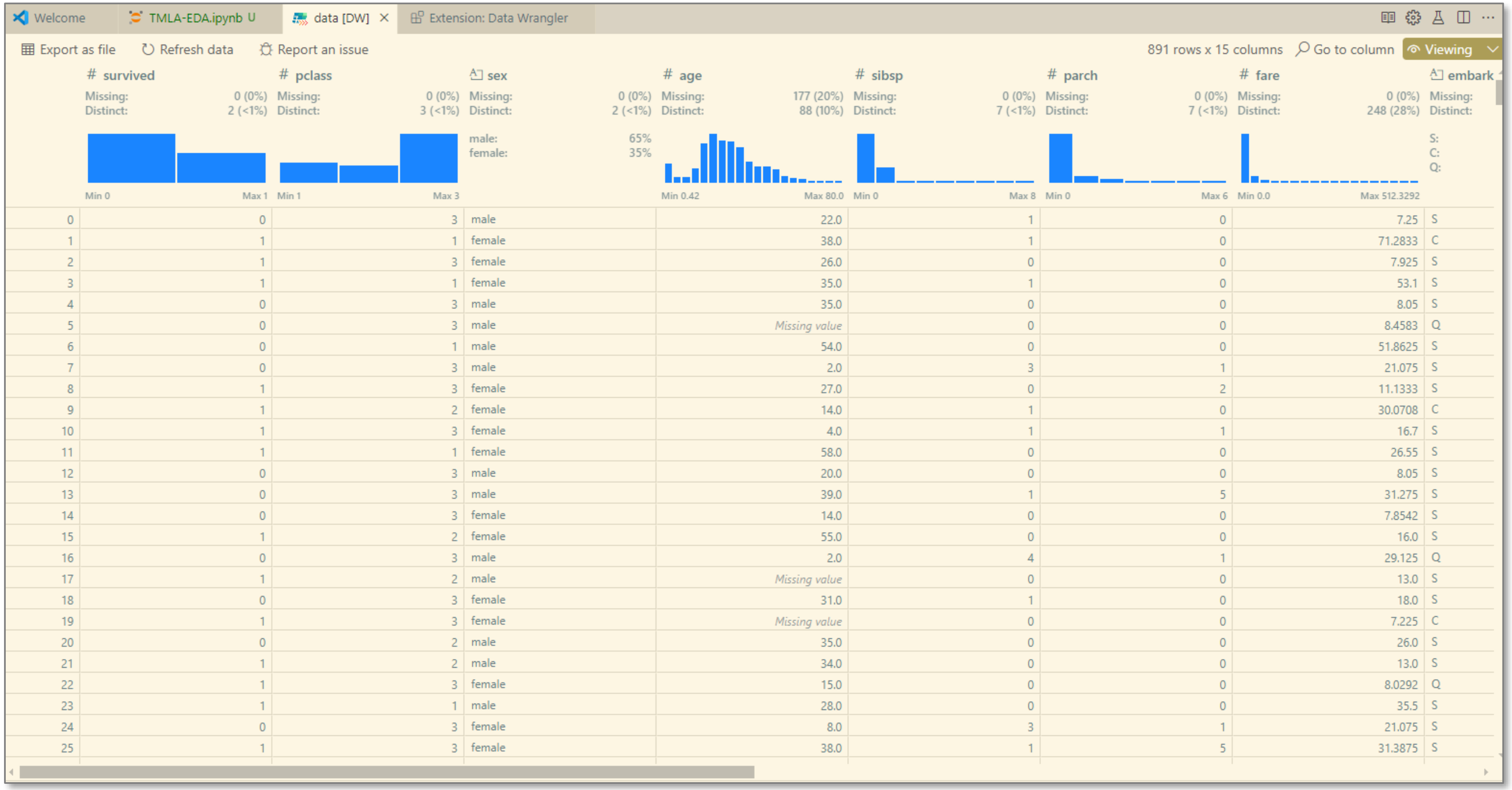
Titanic

Contains information about Titanic passengers, including age, class, and survival (widely used for classification tasks)

Case study of EDA

Titanic dataset

Univariate graphical and non-graphical

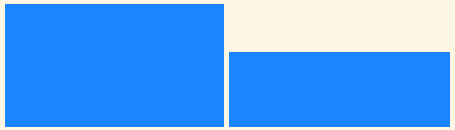
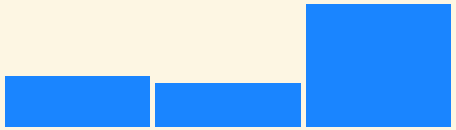
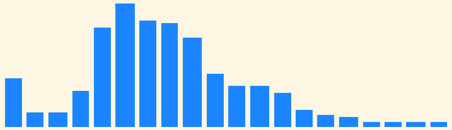


	survived	pclass	sex	age	sibsp	parch	fare	embark
0	0	3	male	22.0	1	0	7.25	S
1	1	1	female	38.0	1	0	71.2833	C
2	1	3	female	26.0	0	0	7.925	S
3	1	1	female	35.0	1	0	53.1	S
4	0	3	male	35.0	0	0	8.05	S
5	0	3	male	Missing value	0	0	8.4583	Q
6	0	1	male	54.0	0	0	51.8625	S
7	0	3	male	2.0	3	1	21.075	S
8	1	3	female	27.0	0	2	11.1333	S
9	1	2	female	14.0	1	0	30.0708	C
10	1	3	female	4.0	1	1	16.7	S
11	1	1	female	58.0	0	0	26.55	S
12	0	3	male	20.0	0	0	8.05	S
13	0	3	male	39.0	1	5	31.275	S
14	0	3	female	14.0	0	0	7.8542	S
15	1	2	female	55.0	0	0	16.0	S
16	0	3	male	2.0	4	1	29.125	Q
17	1	2	male	Missing value	0	0	13.0	S
18	0	3	female	31.0	1	0	18.0	S
19	1	3	female	Missing value	0	0	7.225	C
20	0	2	male	35.0	0	0	26.0	S
21	1	2	male	34.0	0	0	13.0	S
22	1	3	female	15.0	0	0	8.0292	Q
23	1	1	male	28.0	0	0	35.5	S
24	0	3	female	8.0	3	1	21.075	S
25	1	3	female	38.0	1	5	31.3875	S

Case study of EDA

Titanic dataset

Univariate graphical and non-graphical

	# survived	# pclass	sex	# age
	Missing: 0 (0%) Distinct: 2 (<1%)	Missing: 0 (0%) Distinct: 3 (<1%)	Missing: 0 (0%) Distinct: 2 (<1%)	Missing: 177 (20%) Distinct: 88 (10%)
			male: 65% female: 35%	
	Min 0 / Max 1	Min 1 / Max 3		Min 0.42 / Max 80.0
0	0	3	male	22.0
1	1	1	female	38.0
2	1	3	female	26.0
3	1	1	female	35.0
4	0	3	male	35.0
5	0	3	male	*Missing value*
6	0	1	male	54.0
7	0	3	male	2.0
8	1	3	female	27.0
9	1	2	female	14.0

Case study of EDA

Titanic dataset

Univariate non-graphical

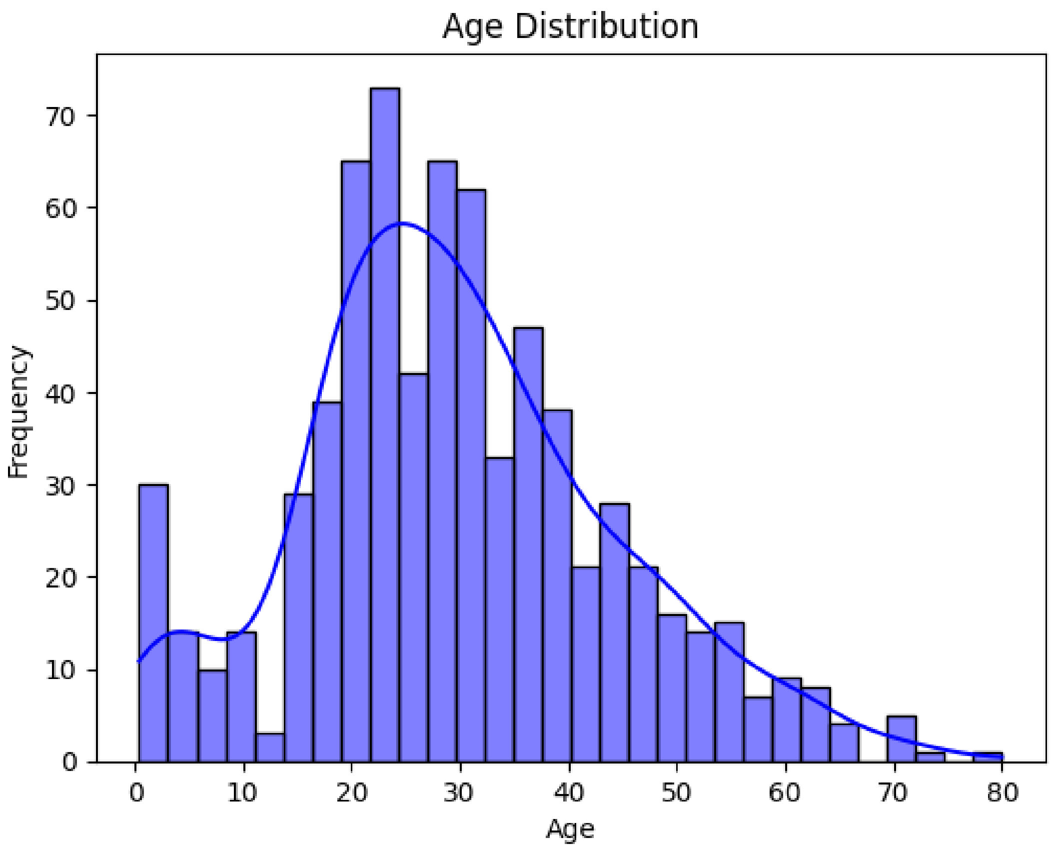
```
Descriptive Statistics:
          survived      pclass   sex         age       sibsp       parch  \
count   891.000000  891.000000   891  714.000000  891.000000  891.000000
unique         NaN         NaN     2         NaN         NaN         NaN
top            NaN         NaN  male         NaN         NaN         NaN
freq           NaN         NaN   577         NaN         NaN         NaN
mean      0.383838    2.308642   NaN   29.699118    0.523008    0.381594
std       0.486592    0.836071   NaN   14.526497    1.102743    0.806057
min       0.000000    1.000000   NaN    0.420000    0.000000    0.000000
25%       0.000000    2.000000   NaN   20.125000    0.000000    0.000000
50%       0.000000    3.000000   NaN   28.000000    0.000000    0.000000
75%       1.000000    3.000000   NaN   38.000000    1.000000    0.000000
max       1.000000    3.000000   NaN   80.000000    8.000000    6.000000

              fare embarked  class  who adult_male deck  embark_town alive  \
count   891.000000      889    891  891        891  203          889   891
unique         NaN        3      3    3          2    7            3     2
top            NaN        S  Third  man       True    C  Southampton    no
freq           NaN      644    491  537        537   59          644   549
mean     32.204208      NaN    NaN  NaN        NaN  NaN          NaN   NaN
std      49.693429      NaN    NaN  NaN        NaN  NaN          NaN   NaN
min       0.000000      NaN    NaN  NaN        NaN  NaN          NaN   NaN
25%       7.910400      NaN    NaN  NaN        NaN  NaN          NaN   NaN
50%      14.454200      NaN    NaN  NaN        NaN  NaN          NaN   NaN
75%      31.000000      NaN    NaN  NaN        NaN  NaN          NaN   NaN
max     512.329200      NaN    NaN  NaN        NaN  NaN          NaN   NaN
```

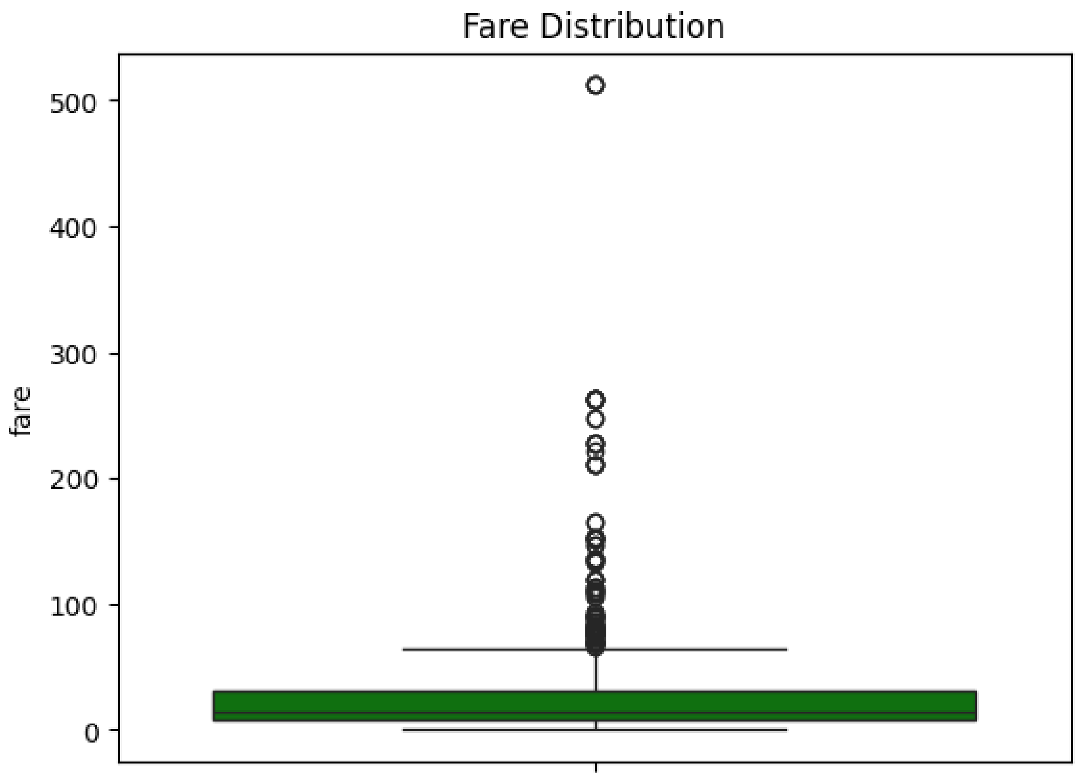
```
       alone
count    891
unique     2
top     True
freq     537
mean     NaN
std      NaN
min      NaN
25%      NaN
50%      NaN
75%      NaN
max      NaN
```

Case study of EDA

Titanic dataset

Univariate graphical

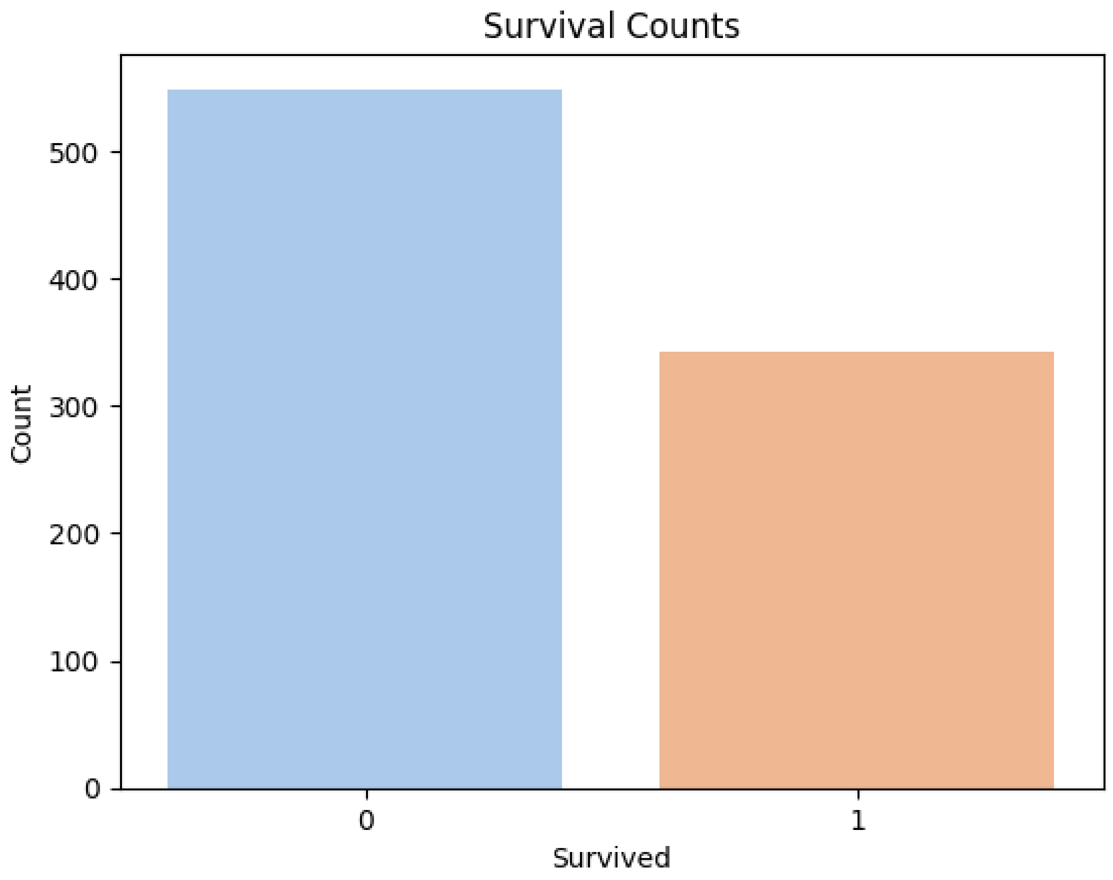




Case study of EDA

Titanic dataset

Univariate graphical



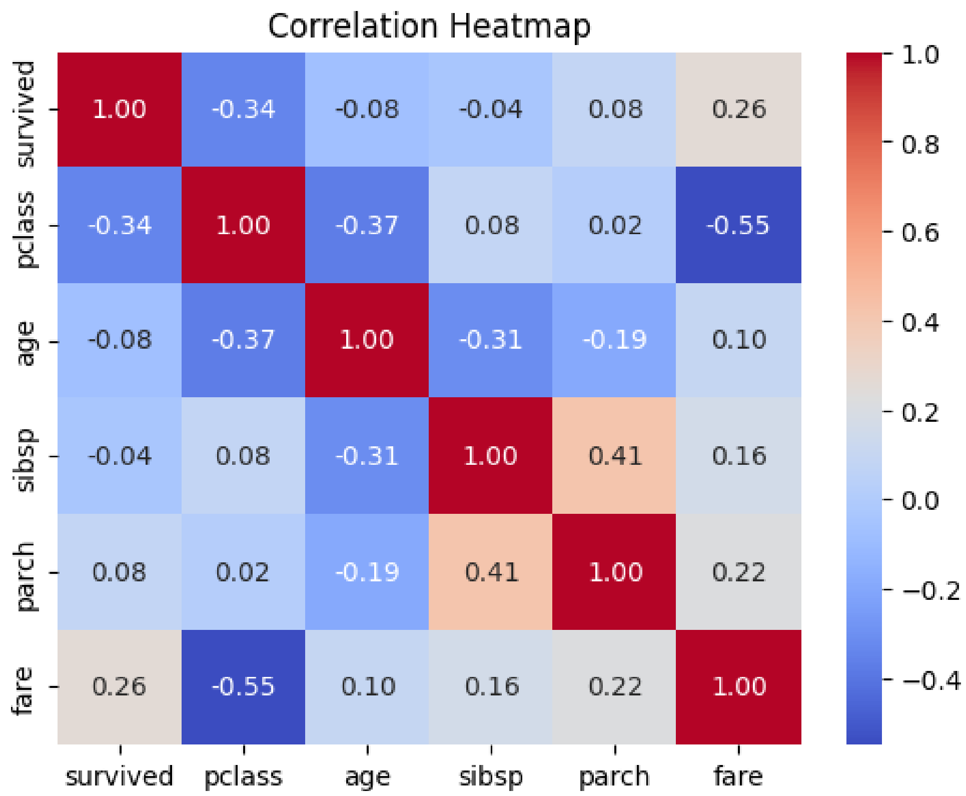
Case study of EDA

Titanic dataset

Multivariate non-graphical

```
Correlation Matrix:
          survived    pclass       age     sibsp     parch      fare
survived  1.000000 -0.338481 -0.077221 -0.035322  0.081629  0.257307
pclass   -0.338481  1.000000 -0.369226  0.083081  0.018443 -0.549500
age      -0.077221 -0.369226  1.000000 -0.308247 -0.189119  0.096067
sibsp    -0.035322  0.083081 -0.308247  1.000000  0.414838  0.159651
parch     0.081629  0.018443 -0.189119  0.414838  1.000000  0.216225
fare      0.257307 -0.549500  0.096067  0.159651  0.216225  1.000000
```

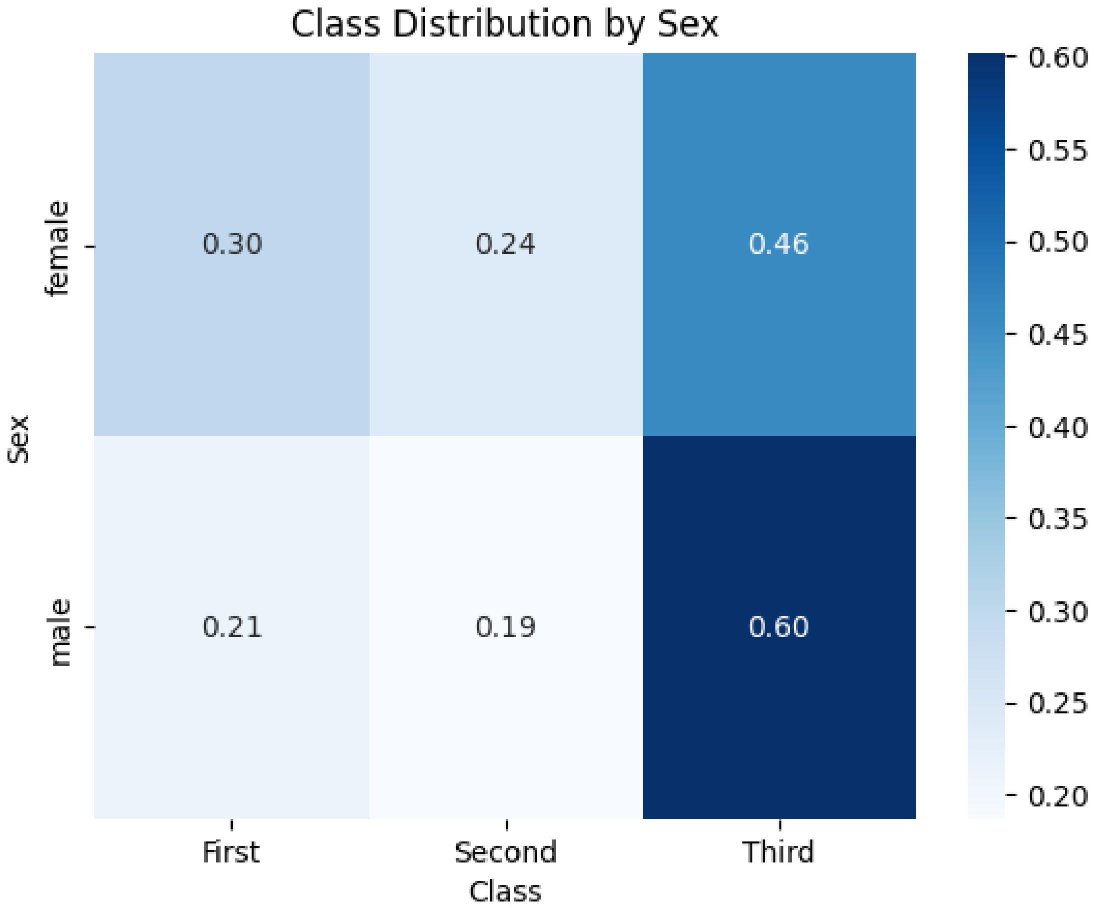
Multivariate graphical



Case study of EDA

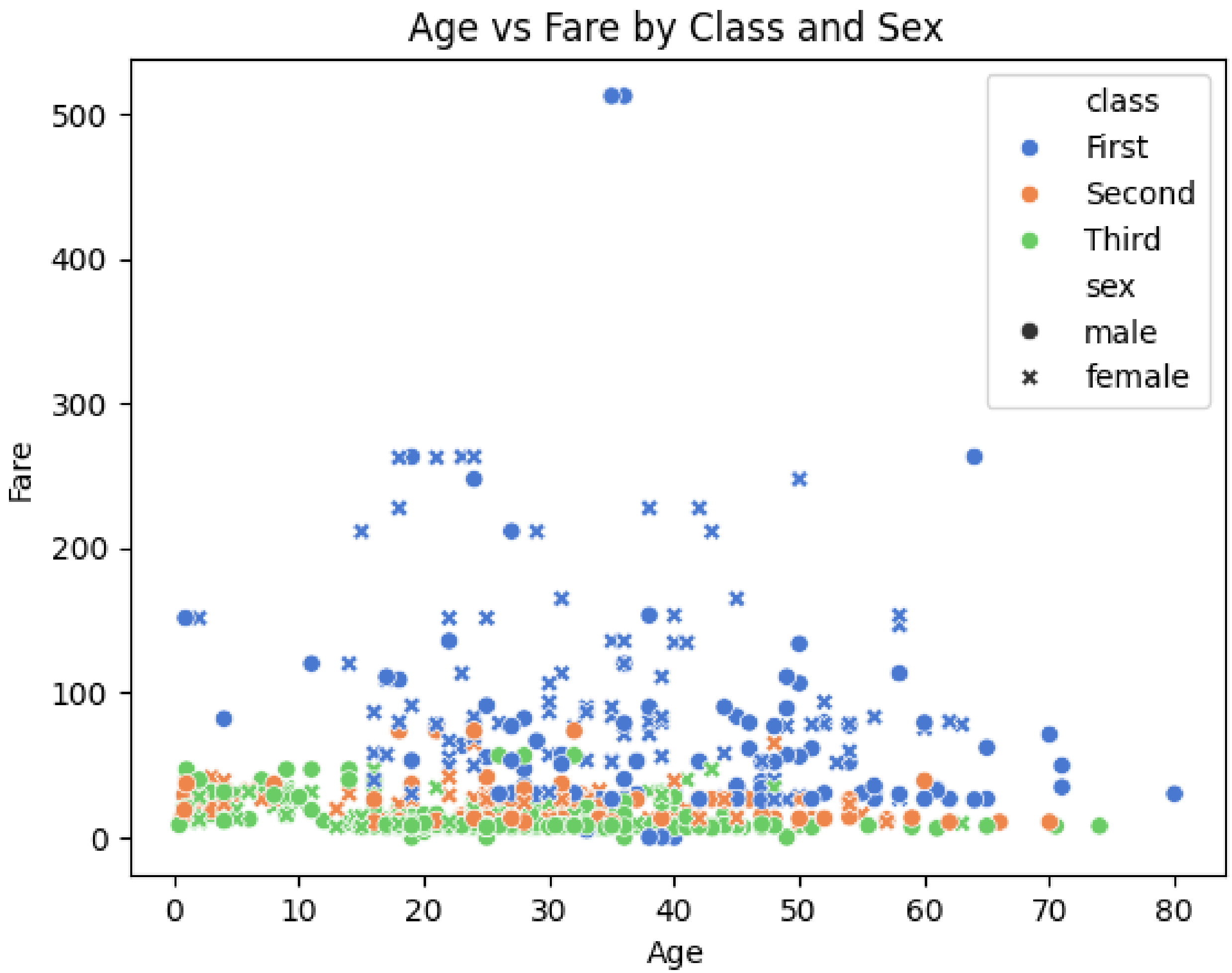
Titanic dataset

Multivariate graphical



Case study of EDA
Titanic dataset

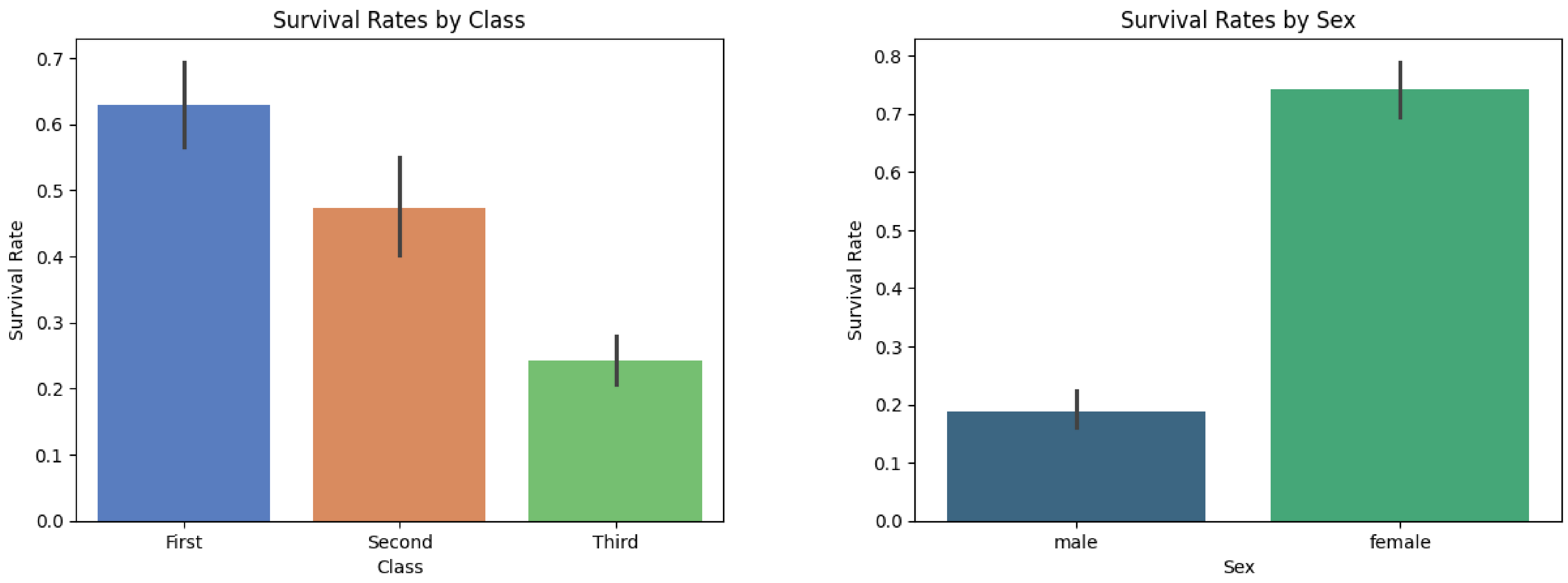
Multivariate graphical



Case study of EDA

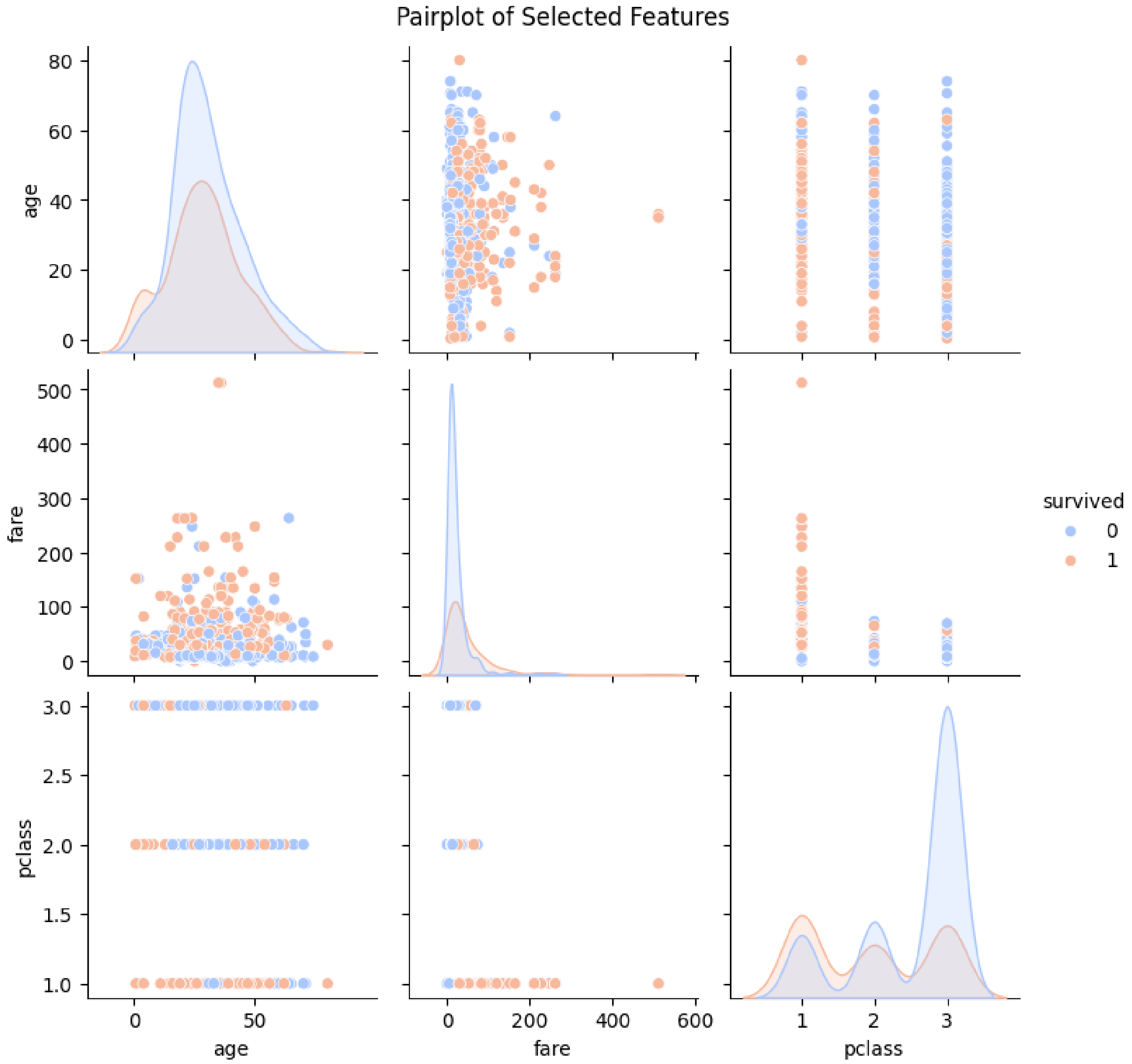
Titanic dataset

Multivariate graphical



Case study of EDA

Titanic dataset

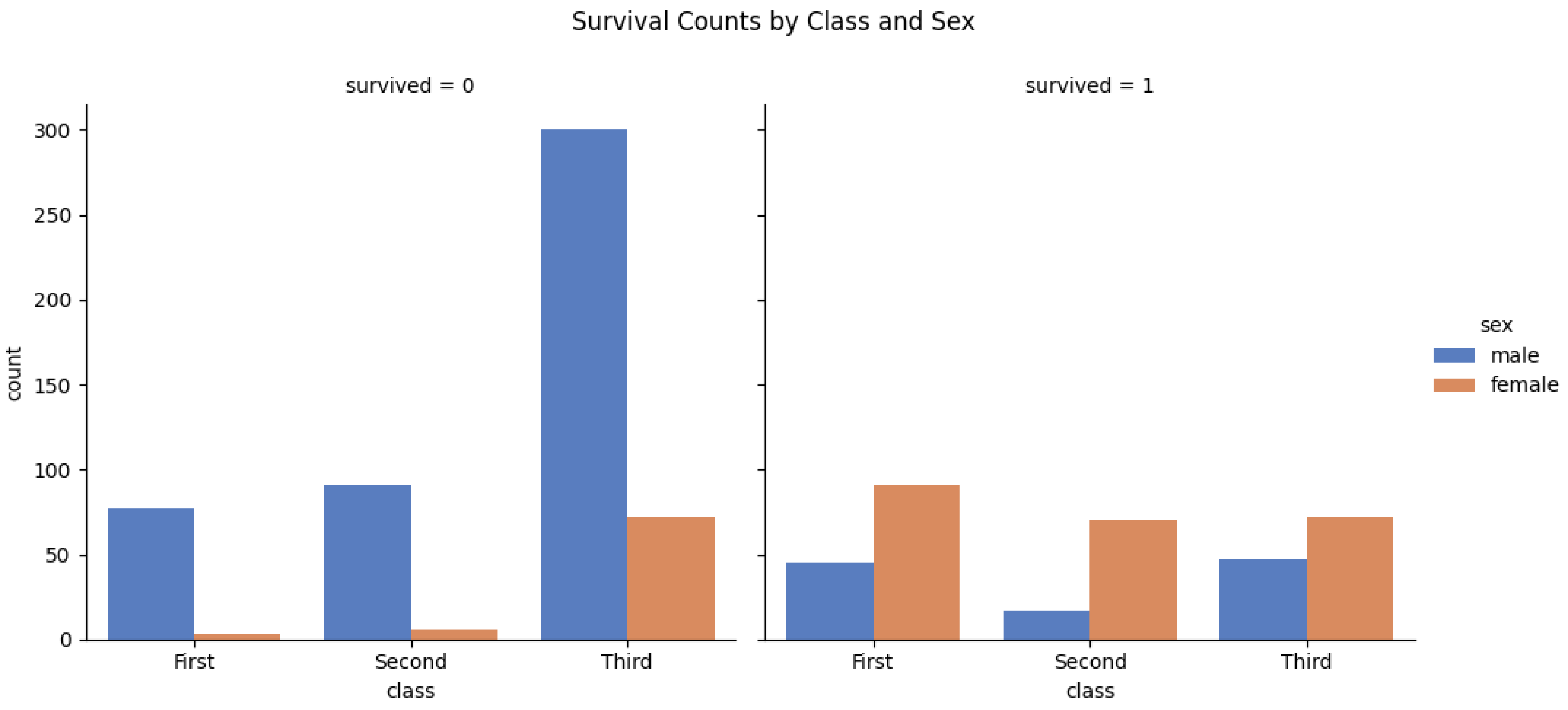


Multivariate graphical

Case study of EDA

Titanic dataset

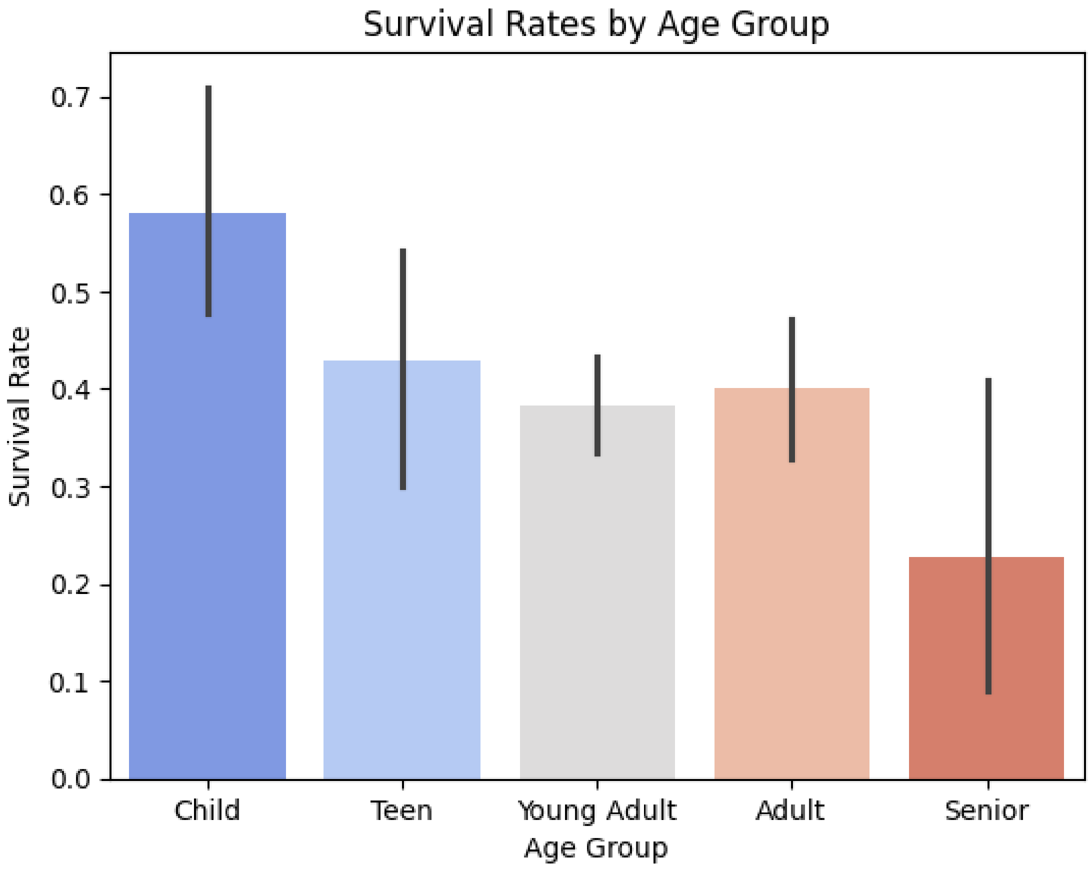
Multivariate graphical



Case study of EDA

Titanic dataset

Multivariate graphical



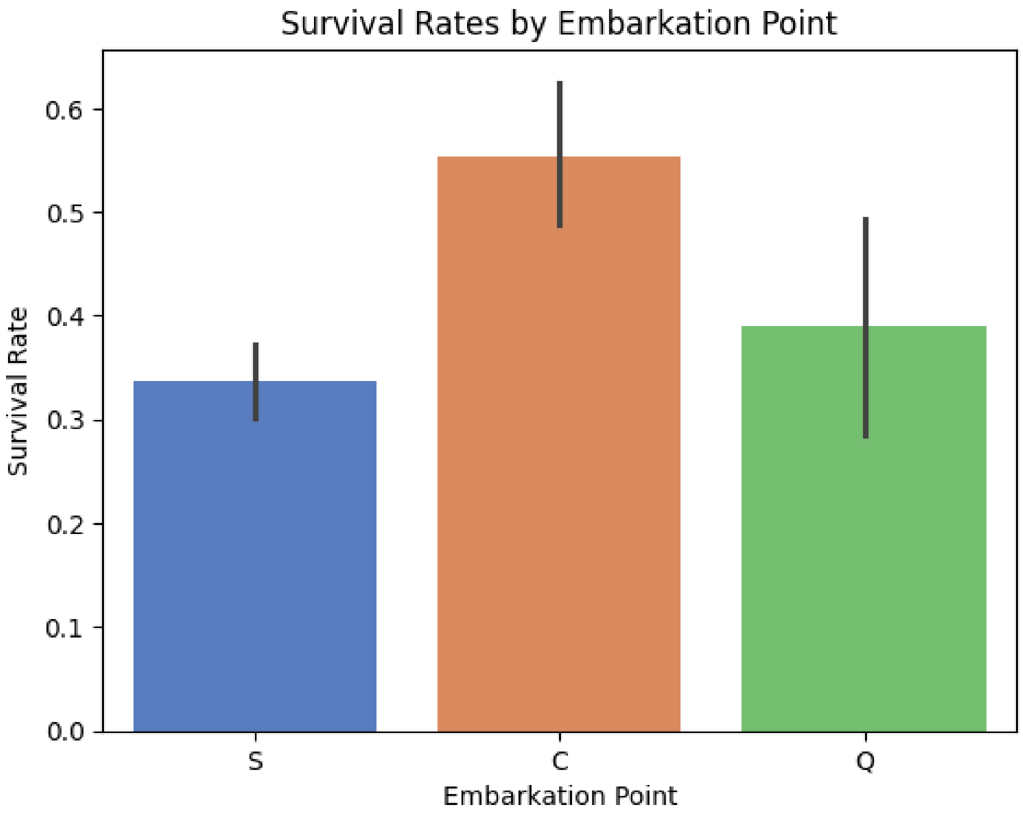
Multivariate non-graphical

```
Survival Rates by Age Group:
age_group
Child          0.579710
Teen           0.428571
Young Adult    0.382682
Adult          0.400000
Senior         0.227273
Name: survived, dtype: float64
```

Case study of EDA

Titanic dataset

Multivariate graphical



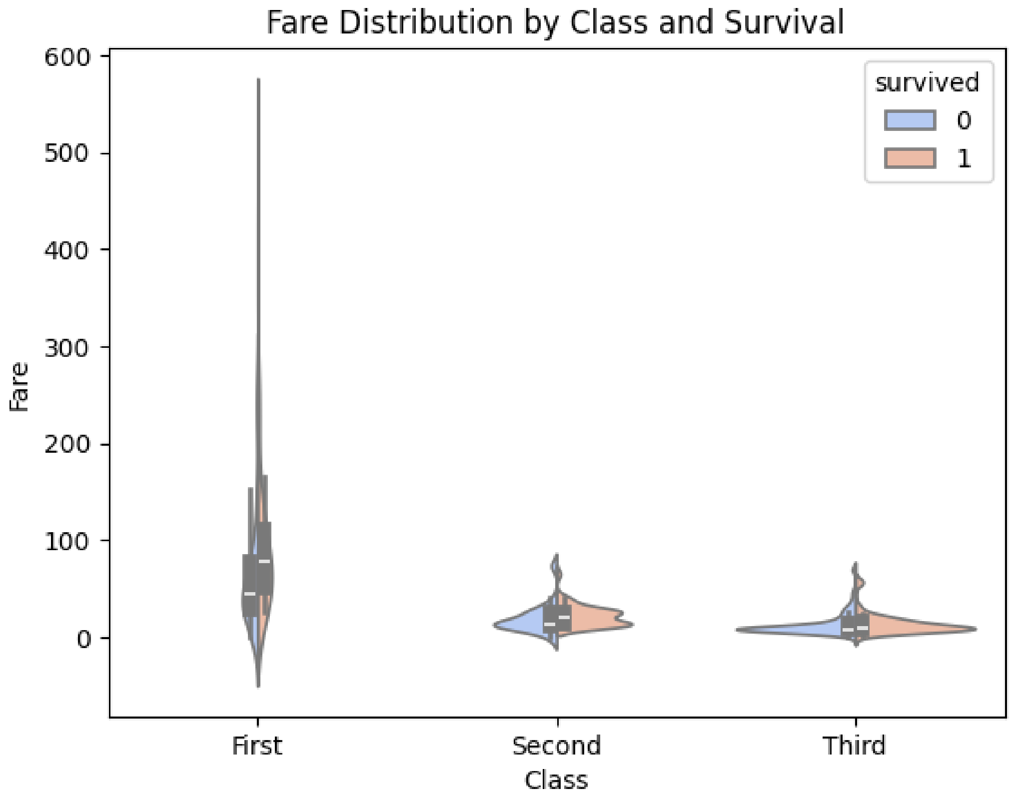
Multivariate non-graphical

```
Survival Rates by Embarkation Point:
embarked
C    0.553571
Q    0.389610
S    0.336957
Name: survived, dtype: float64
```

Case study of EDA

Titanic dataset

Multivariate graphical



Multivariate non-graphical

```
Average Fare by Class and Survival:
class   survived
First   0           64.684007
        1           95.608029
Second  0           19.412328
        1           22.055700
Third   0           13.669364
        1           13.694887
Name: fare, dtype: float64
```

Exploratory data analysis

Data Science Project

- In the context of each project:
  - **Dataset** – What is the dataset? Where is it located?
  - **Goal** – What is the problem you are trying to address? Is it classification or regression?
  - **Data** – What are the independent variables? What's their type?
  - **Challenges** – Which are the issues that need to be addressed when tackling the problem?
  - **Business framing** – How will the model contribute to the business goals?